

Conjoint Analysis

Product characteristics

Attitude formation

Model of choice

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The Attitude

- An attitude is a psychological tendency or disposition that represents an individual's overall
 - evaluation
 - feelings, beliefs,
 - and behavioral intentions
- Toward a particular object, product, brand...
- Attitudes can be positive, negative, or neutral and play a crucial role (purchasing behavior)

Purchasing Behavior

- Routine Purchases (Low Involvement Products)
 - Everyday items that consumers purchase frequently, often without extensive evaluation of attributes.
 - Examples include basic groceries, toiletries, and household cleaning products,...
 - Consumer attitudes toward routine purchases may be more influenced by factors like brand loyalty, past experience, and price promotions.
- Ego-Expressive Products
 - These are products that individuals use to express their identity, style, or personality
 - Examples include luxury fashion items, high-end cosmetics,...
 - Consumers often form strong attitudes toward these products based on how they reflect their self-identity and personal values.
 - Attributes like design, brand image, and exclusivity may be highly important in shaping attitudes and purchase decisions.
- High-Involvement Products
 - Products that require more careful consideration and evaluation due to their cost, complexity, or potential long-term impact.
 - Examples include cars, electronics, and home appliances.
 - Consumers are likely to engage in a systematic evaluation of attributes and form attitudes based on their assessments.

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Attitude formation The Compensatory Model

- For High-Involvement Products
- A compensatory model, for example the Fishbein Model, is a type of decision-making model
 - individuals evaluate a product or brand by considering multiple attributes (beliefs) and assessing the importance of each attribute.
 - They then calculate an overall attitude toward the product by multiplying each belief by its corresponding importance
 - and summing up.

$$A = \sum_{i=1}^k e_i b_i$$

A : attitude towards the product

e_i : Evaluation of the performance in relation to the attribute i

b_i : Importance of the attribute i

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Attitude formation

The Compensatory Model

- Attribute Evaluation
- Importance or weight of each attribute
- Trade-Offs: Consumers can make trade-offs between attributes.
 - lower price but slightly lower quality compared to higher price but better quality
- Overall Evaluation for each alternative
 - helps individuals to rank or choose among the available options
- Applicability
 - for complex purchase decisions involving products or services with multiple attributes, such as cars, smartphones, or vacation packages.

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Why conjoint analysis ?

- Attribute-importance survey :
 - By asking, attribute after attribute, about the importance of this attributes
 - on a scale (for example from 0 to 10)
- Or a Conjoint analysis
 - With a set of products to rank
 - From the most appreciated to the least appreciated

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The importance of product features

A sort of "Rationalization"

Respondents express their views on the general importance of these criteria but not on their decisive nature in their choice

Que penser des méthodes basées sur des questions directes d'importance ?

« Toutes ces méthodes reposent sur deux postulats : les interviewés connaissent l'importance des attributs dans leur choix ou leur achat ; les interviewés acceptent, sans travestir la réalité, de les déclarer à l'enquêteur. Il faut être extrêmement naïf et ignorer des dizaines d'années de recherche psychologique pour pouvoir admettre ces postulats. De plus, il n'est pas fait distinction de l'importance en soi d'un attribut et de son rôle déterminant dans un choix entre marques [...] Dans le domaine automobile les réponses à ce type de questions directes sont "amusantes" comme le montrent les résultats de deux sondages (France) :

« Quelles sont les trois qualités que vous considérez comme les plus importantes pour une automobile ? »

Sécurité active	83 %	Confort	30 %	Esthétique	6 %
Sécurité passive	57 %	Prix d'achat	13 %	Volume intérieur	5 %
Consommation	44 %	Facilité d'entretien	12 %	Vitesse	4 %
Longévité	33 %	Facilité de revente	7 %		

(Source : sondage postal auprès de 1 000 possesseurs d'automobile)

On est frappé par la prédominance des facteurs rationnels (sécurité, économie) sur les facteurs hédonistes comme esthétique et performance ; mais la formulation de la question visait l'importance en soi. Dans l'exemple suivant c'est bien la déterminance qui est recherchée :

« Quels sont les trois critères qui guideraient le choix de votre prochaine voiture ? »			
Prix d'achat	52 %	Économie d'entretien	20 %
Consommation	41 %	Résistance à l'usure	18 %
Sécurité, confort	29 %	Réputation marque	17 %
Habitabilité	25 %	Carrosserie, ligne	14 %
		Vitesse	12 %
		Puissance	11 %

(Source : Sondage réalisé auprès de 2 800 automobilistes)

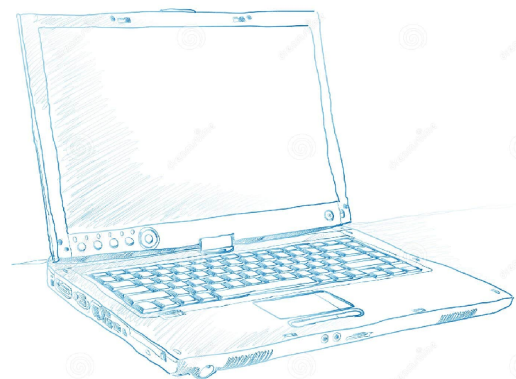
Même sans être spécialiste du marketing automobile, vous trouverez peu plausible que l'esthétique ne soit qu'en 7^e position et soit considérée comme trois fois moins importante que la consommation de carburant. Vous aurez raison ! Dans la même enquête, le modèle largement en tête des choix est la [Peugeot] 205 dont le prix et la consommation ne sont pas les points forts perçus, au contraire de la ligne et des performances.

Il faut donc malheureusement conclure qu'une partie importante des interviewés, soit ne connaît pas sa hiérarchie d'attente, soit rationalise ses réponses, soit encore ne répond pas à une question sur le choix (déterminance), mais à une question sur l'importance en soi. Et il faut avoir le courage de dire que l'utilisation de ces méthodes simplistes pour mesurer l'importance des attentes de la clientèle dans les choix jette le discrédit sur la recherche en marketing [...].

Source : Bachelet D. et Lion J., « Une méthode d'évaluation de l'importance des attributs perçus appliquée au développement et au positionnement des nouveaux produits », *Revue Française du Marketing*, n° 119, 1988/4, p. 5-26.

Conjoint analysis

- 12 computer configurations
- Price:
 - Low (\$500)
 - Medium (\$600)
 - High (\$700)
- Screen Size:
 - Small (15 inches)
 - Large (17 inches)
- Hard drive Size:
 - Small (1To)
 - Large (2To)



- 3 characteristics
- 12 combinations = 3x2x2
- Which one should be proposed?

Conjoint analysis

- 12 computer configurations
- 12 combinations of 3 characteristics



	Screen	HD	Price
P1	15"	HD 1 to	€700.00
P2	15"	HD 1 to	€600.00
P3	15"	HD 1 to	€500.00
P4	15"	HD 2 to	€700.00
P5	15"	HD 2 to	€600.00
P6	15"	HD 2 to	€500.00
P7	17"	HD 1 to	€700.00
P8	17"	HD 1 to	€600.00
P9	17"	HD 1 to	€500.00
P10	17"	HD 2 to	€700.00
P11	17"	HD 2 to	€600.00
P12	17"	HD 2 to	€500.00

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Why conjoint analysis ?

- Realistic modeling of the decision-making process
 - simulates a realistic decision-making process in which consumers face complex choices between several products or services
 - possible to gather more precise information on how consumers actually make purchasing decisions.
 - This better reflects the reality where consumers often have to choose between options with advantages and disadvantages.
- Identification of latent preferences
 - Conjoint analysis makes it possible to discover the latent preferences of consumers (what they really appreciate) even if they do not express it clearly in a simple rating survey.

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Conjoint analysis

- Study of the influence of object characteristics on the classification of these objects.
- Basic data and assumptions
 - Variable to be explained: Preference
 - Explanatory variables: The characteristics of the products described using a set of variables
- Interpretation: The overall goodness of fit reflects the relevance of features in establishing consumer choice
- Result: Relative importance of attributes



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Conjoint analysis

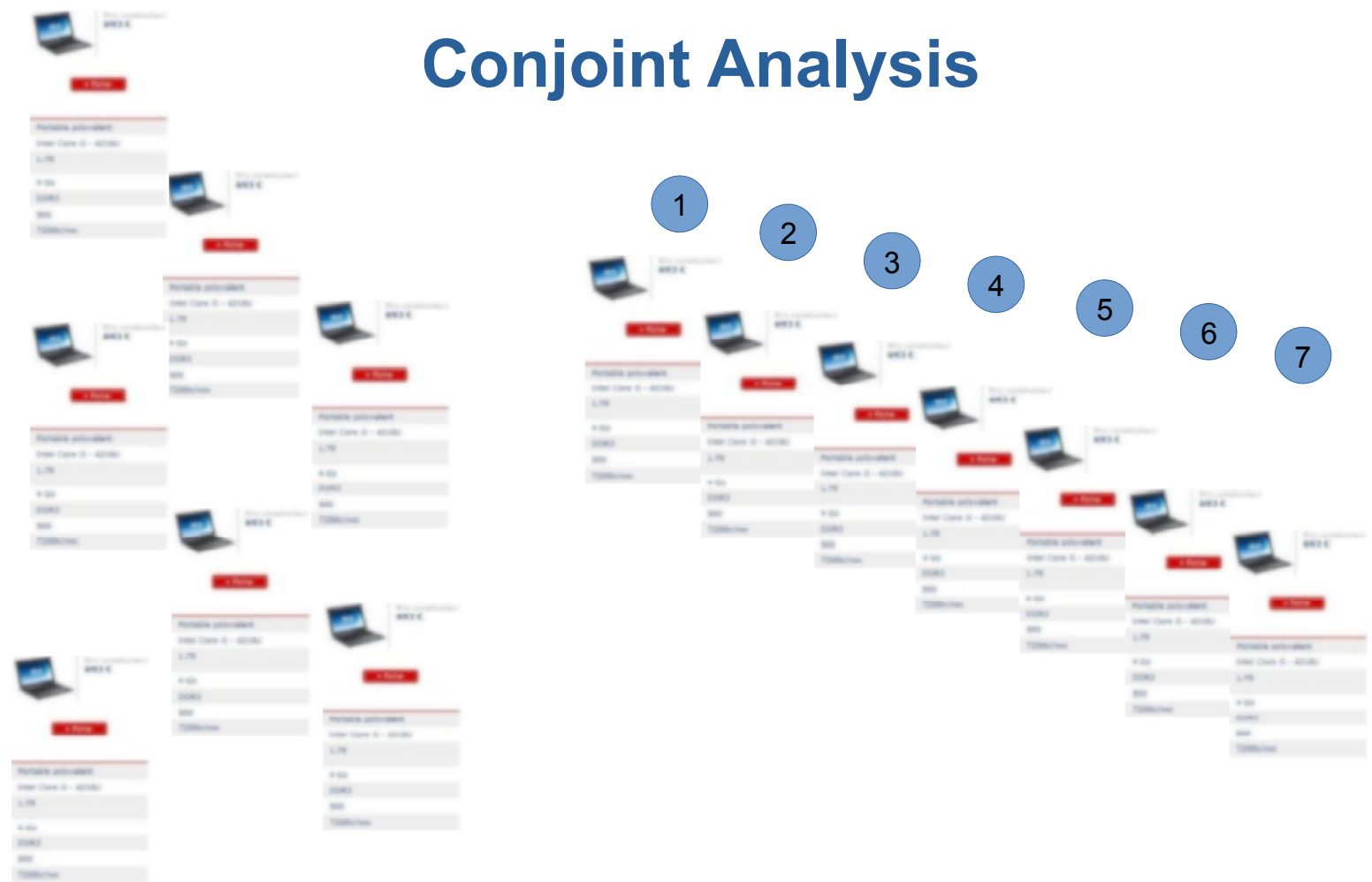
- 12 computer configurations
- 12 combinations of 3 characteristics



	Screen	HD	Price
P1	15"	HD 1 to	€700.00
P2	15"	HD 1 to	€600.00
P3	15"	HD 1 to	€500.00
P4	15"	HD 2 to	€700.00
P5	15"	HD 2 to	€600.00
P6	15"	HD 2 to	€500.00
P7	17"	HD 1 to	€700.00
P8	17"	HD 1 to	€600.00
P9	17"	HD 1 to	€500.00
P10	17"	HD 2 to	€700.00
P11	17"	HD 2 to	€600.00
P12	17"	HD 2 to	€500.00

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Conjoint Analysis



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Application Conjoint (or similar) analysis

	Config	Ecran_15	Ecran_17	HD_1To	HD_2To	Prix_700	Prix_600	Prix_500	Ind1	Ind2	Ind3	Ind4	Ind5	Ind6	Ind7	Ind8
1	P1	1	0	1	0	1	0	0	12	10	11	12	12	12	11	
2	P2	1	0	1	0	0	1	0	8	11	10	8	10	8	9	
3	P3	1	0	1	0	0	0	1	4	12	12	4	11	4	4	
4	P4	1	0	0	1	1	0	0	11	6	6	11	6	10	12	
5	P5	1	0	0	1	0	1	0	7	4	4	6	4	7	7	
6	P6	1	0	0	1	0	0	1	3	5	5	3	5	3	3	
7	P7	0	1	1	0	1	0	0	9	9	9	9	9	9	10	
8	P8	0	1	1	0	0	1	0	6	7	7	7	7	6	6	
9	P9	0	1	1	0	0	0	1	2	8	8	2	8	2	2	
10	P10	0	1	0	1	1	0	0	10	3	3	10	3	11	8	
11	P11	0	1	0	1	0	1	0	5	1	1	5	1	5	5	
12	P12	0	1	0	1	0	0	1	1	2	2	1	2	1	1	

- 12 Computer Configurations Ranked in Market Research study by 36 Consumers
- The conjoint analysis makes it possible to evaluate the interest of a characteristic relatively (jointly) to the others (“Big hard disk” or “lower price”)

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Application

Conjoint (or similar) analysis

	Config	Ecran_15	Ecran_17	HD_1To	HD_2To	Prix_700	Prix_600	Prix_500
1	P1	1	0	1	0	1	0	0
2	P2	1	0	1	0	0	1	0
3	P3	1	0	1	0	0	0	1
4	P4	1	0	0	1	1	0	0
5	P5	1	0	0	1	0	1	0
6	P6	1	0	0	1	0	0	1
7	P7	0	1	1	0	1	0	0
8	P8	0	1	1	0	0	1	0
9	P9	0	1	1	0	0	0	1
10	P10	0	1	0	1	1	0	0
11	P11	0	1	0	1	0	1	0
12	P12	0	1	0	1	0	0	1

Step 1: Each consumer surveyed ranks the 12 configurations in descending order of their preferences (Extract below)

	Config	Ecran_15	Ecran_17	HD_1To	HD_2To	Prix_700	Prix_600	Prix_500	Ind1	Ind2	Ind3	Ind4	Ind5	Ind6	Ind7	Ind8
1	P1	1	0	1	0	1	0	0	12	10	11	12	12	12	11	
2	P2	1	0	1	0	0	1	0	8	11	10	8	10	8	9	
3	P3	1	0	1	0	0	0	1	4	12	12	4	11	4	4	
4	P4	1	0	0	1	1	0	0	11	6	6	11	6	10	12	
5	P5	1	0	0	1	0	1	0	7	4	4	6	4	7	7	
6	P6	1	0	0	1	0	0	1	3	5	5	3	5	3	3	
7	P7	0	1	1	0	1	0	0	9	9	9	9	9	9	10	
8	P8	0	1	1	0	0	1	0	6	7	7	7	7	6	6	
9	P9	0	1	1	0	0	0	1	2	8	8	2	8	2	2	
10	P10	0	1	0	1	1	0	0	10	3	3	10	3	11	8	
11	P11	0	1	0	1	0	1	0	5	1	1	5	1	5	5	
12	P12	0	1	0	1	0	0	1	1	2	2	1	2	1	1	

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Application

Conjoint (or similar) analysis

Step 2: We transform the rows into points...

```
# Extract the ranks data from columns 12 to 47 in the Excel sheet
Ranks <- data[12:47]

# *****
# Simplified Approach
# *****

# Transform rank values into points obtained (higher rank = fewer points)
Pts <- 13 - Ranks

# Calculate the mean points for each configuration
meansPts <- rowMeans(Pts)
print(meansPts)
```

12 points for the config ranked 1

11 ... 2

10 ... 3

13 - rank achieved

Step 3: Calculate the average number of points obtained by each configuration

Ind1	Ind2	Ind3	Ind4	Ind5	Ind6	Ind7	Ind8
12	10	11	12	12	12	11	
8	11	10	8	10	8	9	
4	12	12	4	11	4	4	
11	6	6	11	6	10	12	
7	4	4	6	4	7	7	
3	5	5	3	5	3	3	
9	9	9	9	9	9	10	
6	7	7	7	7	6	6	
2	8	8	2	8	2	2	
10	3	3	10	3	11	8	
5	1	1	5	1	5	5	
1	2	2	1	2	1	1	

```
> print(meansPts)
[1] 1.333333 4.222222 6.666667 3.638889 6.888889 9.388889 3.361111 6.722222 9.111111 5.777778
[11] 9.194444 11.694444
```

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Application

Conjoint (or similar) analysis

Step 4: Linear regression is used to explain variations in the number of points by the presence or absence of a characteristic.

We give ourselves a basic configuration, here 15" screen, HD 1TB, Price 700€ (problem of multicollinearity)

meansPts	Config	Ecran_15	Ecran_17	HD_1To	HD_2To	Prix_700	Prix_600	Prix_500
1.333333	P1	1	0	1	0	1	0	0
4.222222	P2	1	0	1	0	0	1	0
6.666667	P3	1	0	1	0	0	0	1
3.638889	P4	1	0	0	1	1	0	0
6.888889	P5	1	0	0	1	0	1	0
9.388889	P6	1	0	0	1	0	0	1
3.361111	P7	0	1	1	0	1	0	0
6.722222	P8	0	1	1	0	0	1	0
9.111111	P9	0	1	1	0	0	0	1
5.777778	P10	0	1	0	1	1	0	0
9.194444	P11	0	1	0	1	0	1	0
11.694444	P12	0	1	0	1	0	0	1

$$Pts = \beta_0 + \beta_{ec17} \cdot VB_{ec17} + \beta_{hd2t} \cdot VB_{hd2t} + \beta_{500\text{€}} \cdot VB_{500\text{€}} + \beta_{600\text{€}} \cdot VB_{600\text{€}}$$

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Application: Conjoint Analysis Results

Call:

```
lm(formula = meansPts ~ data_ac$Ecran_17 + data_ac$HD_2To + data_ac$Prix_500 + data_ac$Prix_600)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.15741	-0.06655	0.01389	0.05787	0.21296

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	1.12037	0.08630	12.98	3.74e-06 ***
data_ac\$Ecran_17	2.28704	0.07719	29.63	1.29e-08 ***
data_ac\$HD_2To	2.52778	0.07719	32.75	6.41e-09 ***
data_ac\$Prix_500	5.68750	0.09454	60.16	9.20e-11 ***
data_ac\$Prix_600	3.22917	0.09454	34.16	4.78e-09 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.1337 on 7 degrees of freedom
Multiple R-squared: 0.9987, Adjusted R-squared: 0.998
F-statistic: 1398 on 4 and 7 DF, p-value: 3.107e-10

For an equivalent cost, is it more interesting to:

- Propose a *larger screen or HD* ?
- Offer a big screen and a big HD or a *lower price of 200€* ?
-

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Notes about Conjoint Analysis application

- Specific package “Conjoint” in the R language
- Too much product profiles (combinations of characteristics)
 - Experiment design to work with less profiles without results reliability loss
 - Function “caFactorialDesign”
- Ordinal values taken for metric ones
 - Conjoint Analysis try for each respondent to respect his given ranking

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Why conjoint analysis: other advantages

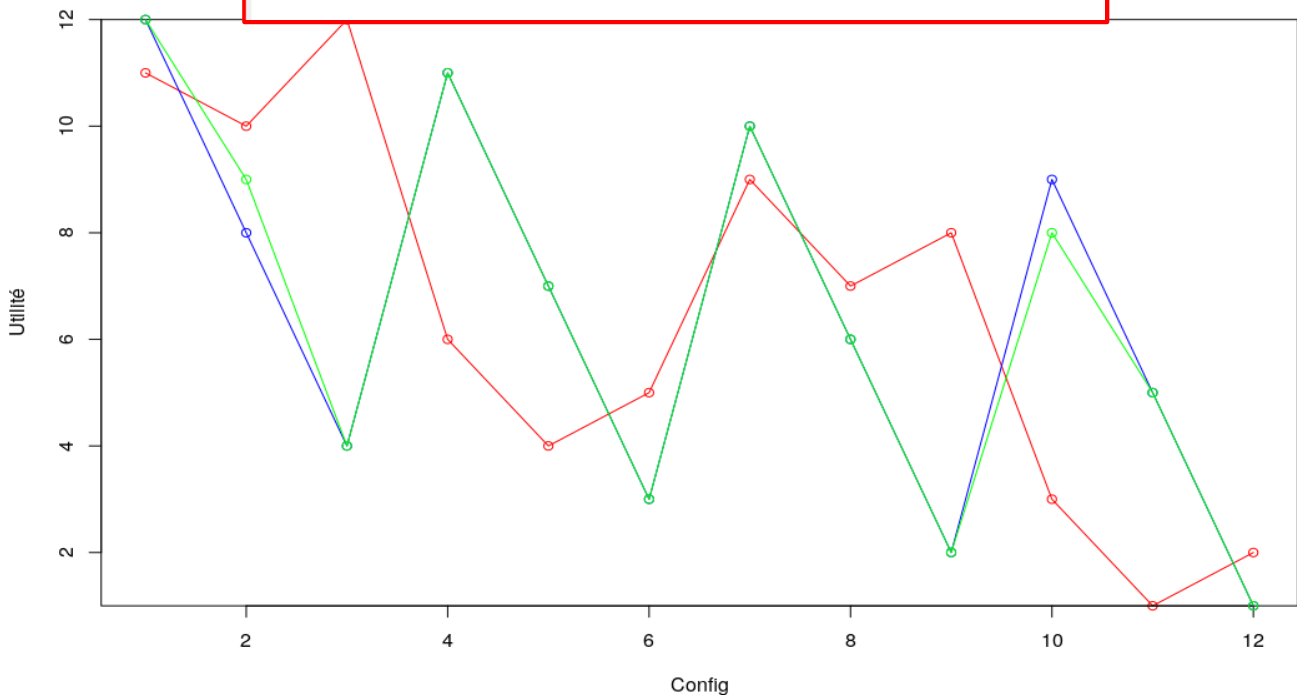
- Saturation effect management: Biased responses when all attributes are considered important.
- Analysis of the interaction between attributes: Conjoint analysis allows analysis of how attribute levels interact with each other.
- Consumer segmentation: Conjoint analysis can be used to segment consumers into groups

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Application: Conjoint analysis

Continuation and end

```
# Pourquoi segmenter ? une évidence graphique  
# Représentation du jugement de 3 individus  
plot(data_ac$Ind21,type = "o",col = "red",ylab="Utilité",xlab="Config" )  
lines(data_ac$Ind8, type = "o", col = "blue")  
lines(data_ac$Ind17, type = "o", col = "green")
```



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Simplified and “traditional” conjoint approach

- Compare the results obtained using the two approaches. What do you think ?
- Use the function “caFactorialDesign” to reduce the number of alternatives to compare.
- Conduct a conjoint analysis with data from you.

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